MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2021

Mathematical Methods

Module Leader: Dr Tim Jupp

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination

SECTION A

1. (a) Find the critical point of

$$f(x, y) = 2xy - 3y,$$

and determine whether it is a local minimum, a local maximum, or a saddle point. (10)

(b) Express the definite integral

$$\int_2^3 (3x - 1)\, dx$$

as a Riemann sum and hence evaluate it. Verify your answer by calculating the integral using the rules of integration. (5)

(c) Evaluate the following definite integrals by using suitable substitutions:

(i)

$$\int_0^1 \frac{t\, dt}{\sqrt{1 + t^2}},$$

(ii)

$$\int_0^5 x^2\sqrt{25 - x^2}\, dx.$$

(10)

(d) Using Gaussian elimination, find the inverse (if it exists) of

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \\ -2 & 1 & 1 \end{pmatrix}.$$

(10)

(e) Solve the ordinary differential equations:

(i)

$$\frac{dy}{dx} + \frac{3}{x}y = 4 + x^2 \quad \text{with } y(1) = 0,$$

(ii)

$$\frac{d^2y}{dx^2} + y = e^{-x} \quad \text{with } y(0) = -\frac{1}{2},\ y'(0) = \frac{1}{2}.$$

(10)

(f) Suppose that $f(x) = 2x^2 - x + 3$ and consider

$$A = \begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix}.$$

Find the matrix $f(A)$. (5)

[50]

SECTION B

2. (a) Evaluate

$$I = \iint_D (|x| + |y|)\ dx\ dy,$$

where D is the region defined by $x^2 + \frac{1}{4}y^2 \leq 1$. (13)

(b) Suppose that $f(x)$ is continuous on $[0, 1]$ and that

$$\int_0^1 f(x)\ dx = \sqrt{3},$$

and hence find the numerical value of the integral

$$\int_0^1 \int_x^1 f(x)f(y)\ dy\ dx.$$

(12)

[**25**]

3. (a) Let $f(x, y) = 2x^2 + 4x + 3y + 2y^2 - 3$.

(i) Let $k > 0$ be a real number. In terms of k, find the points (x, y) for which $f(x, y)$ attains local extrema, subject to the constraint $x^2 + y^2 = k^2$. At each local extremum, find the value of f and state whether this is a local minimum, a local maximum, or neither.

(ii) What is the maximum value that $f(x, y)$ takes in the region $x^2 + y^2 \leq 4$? Justify your answer. (13)

(b) Suppose that $g(x, y)$ satistifes the equation

$$x^2 \frac{\partial^2 g}{\partial x^2} - y^2 \frac{\partial^2 g}{\partial y^2} = 0.$$

Let $u(x, y) = xy$ and $v(x, y) = x/y$. Show that the equation above can then be reduced to

$$\frac{\partial}{\partial v}\left(\frac{\partial g}{\partial u} - \frac{g}{2u}\right) = 0.$$

(12)

[**25**]

4. (a) Use row operations to put the following system of linear equations into reduced row echelon form:

$$\begin{aligned} w + 3x + 3y + z &= 10, \\ 2w + x - y &= 1, \\ 3w - 2x + y + 6z &= 33. \end{aligned}$$

Hence, find all solutions to the system (if any exist). If there are infinitely many solutions, you should give the general solution, indicating clearly which values may be chosen arbitrarily. (12)

(b) For $n \times n$ matrices A and B determine whether each of the following statements is always true or can be false. Justify your answers by giving a proof or a counterexample. (The notation M^T denotes the transpose of a matrix M and I is the $n \times n$ identity matrix.)

(i) If A^3 is the zero matrix, then $\det(A) = 0$.

(ii) $(A^T B)^T = AB^T$.

(iii) $AA^T = A^T A$.

(iv) If $A^2 = 0$ then $I + A$ is invertible.

(13)

[**25**]